

Module 02: Agents in Robotics

Learning Objectives

1. Define **Agents** within the context of Robotics.
2. Analyze how Agents interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Agents.

Introduction

This module explores **Agents**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Agents is a critical component of the 8-part Active Inference spine, bridging the gap between Systems and Perception.

Key Concepts

1. Agents as a Markov Blanket Boundary

How does Agents define the boundary between the agent and the environment?

2. Generative Models of Agents

What parameters involved in Agents must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Agents drive the perception-action loop?

Applications

In Robotics, we see Agents manifest in: * **Specific Example 1:** A fully autonomous delivery drone (such as a Wing or Zipline platform) operates as an Active Inference agent whose

Markov blanket encompasses GPS/INS sensors and optical flow cameras (sensory states) and rotor speed commands plus payload release mechanisms (active states); the drone's onboard generative model encodes flight dynamics, weather predictions, and delivery zone geometry, and it autonomously navigates, avoids obstacles, and executes precision payload drops by continuously minimizing expected free energy without any human teleoperation. * **Specific Example 2:** In a multi-agent autonomous warehouse (such as Amazon Robotics), each Kiva-style mobile robot functions as an independent Active Inference agent that maintains beliefs about its own position, neighboring robots' predicted trajectories, and shelf inventory states; the agent's Markov blanket separates its local decision-making from the global warehouse state, and emergent coordination arises because each agent's generative model includes predictions of other agents' likely actions, enabling collision-free navigation through shared aisles without centralized traffic control.

Conclusion

Understanding Agents allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.